Zhixian Zhang

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Education

Rutgers University New Jersey, USA
Ph.D. in Statistics 2025.09 – present
- Advisor: Linjun Zhang

Zhejiang University Hangzhou, China
Bachelor of Science, Statistics 2021.09 – 2025.06
- GPA: 3.83/4.0

Research Interest

Statistical foundations of AI; AI safety (LLM watermarking); statistically grounded LLM evaluation; machine learning theory; semiparametric inference; optimization.

Research Experience

Evaluating LLMs When They Do Not Know the Answer 2025.10 – 2026.01

- Studied LLM evaluation as a **statistical estimation problem** and developed semiparametric methods with **comparison signals** to assess model capabilities under limited ground-truth labels, leveraging the generation–verification gap.
- Derived an **Efficient Influence Function (EIF)-based one-step estimator** with cross-fitting, establishing asymptotic normality, semiparametric efficiency, and **strict variance reduction** over naive accuracy estimation.
- Validated the method through simulations and **small-sample experiments** on GPQA Diamond (N=50), AIME 2025 (N=15), and GSM8K (N=100), **reducing absolute estimation error** by up to 5.6, 12.0, and 3.5 percentage points, respectively, compared with the naive estimator.

Unified Inference Framework for Single and Multi-Player Performative Prediction 2025.01 – 2026.05

- Developed a **unified inference framework of performative prediction** in both single- and multi-player setting, enabling uncertainty quantification for performative stability and optimality.
- Proposed **repeated-risk-minimization** based on Monte Carlo estimator and fixed point theorem and **two-step plug-in estimators** based on surrogate model theory and importance sampling, and proved **central limit theorems and semiparametric efficiency bounds** for both performative stability and optimality under distributional feedback.

Preprints/Publications

- Conference Paper
  - Zihan Dong, **Zhixian Zhang**, Yang Zhou, Can Jin, Ruijia Wu, Linjun Zhang (2026). Evaluating LLMs When They Do Not Know the Answer: Statistical Evaluation of Mathematical Reasoning via Comparative Signals. ICML 2026. arXiv:2602.03061
- Journal Paper
  - **Zhixian Zhang**, Xiaotian Hou, and Linjun Zhang (2026). Unified Inference Framework for Single and Multi-Player Performative Prediction: Method and Asymptotic Optimality. To be submitted to *The Annals of Statistics*. arXiv:2602.03049

Technical Skills

- Coding Language: Python, R, SQL, Matlab, LaTeX
- Libraries and Tools: PyTorch, TensorFlow, Scikit-Learn, Pandas, Numpy, Scipy, Matplotlib, etc.

Selected Honors & Awards

- Outstanding Student of Zhejiang University 2025.11
- Meritorious Winner, International Mathematical Contest in Modeling(MCM) 2023.02